OS & SP

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SKILL – 03

1.

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GNU nano 8.7.1 skill3.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_LENGTH 100

int main() {
    char history[MAX_HISTORY][MAX_LENGTH];
    char input[MAX_LENGTH];
    int count = 0;
    int current = -1;

    printf("=== Command History Program ===\n");
    printf("Enter commands (type 'exit' to stop)\n\n");

    while (1) {
        printf("Command: ");
        fgets(input, MAX_LENGTH, stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (count < MAX_HISTORY) {
            strcpy(history[count], input);
            count++;
        } else {
            for (int i = 1; i < MAX_HISTORY; i++)
                strcpy(history[i - 1], history[i]);

            strcpy(history[MAX_HISTORY - 1], input);
        }

        current = count;
```

```
        }

        current = count;

        printf("Command stored successfully.\n");
    }

    printf("\n=== Command History ===\n");

    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
    }

    printf("\nEscape Sequence Demonstration:\n");
    printf("New Line -> Hello\nWorld\n");
    printf("Tab -> Hello\tWorld\n");
    printf("Backslash -> C:\\Program Files\n");

    return 0;
}
```

Output :

```
sadwika@DESKTOP-8CQAO1L:~$ mkdir Skill
mkdir: Skill: File exists
sadwika@DESKTOP-8CQAO1L:~$ cd Skill
sadwika@DESKTOP-8CQAO1L:~/Skill$ nano skill3.c
sadwika@DESKTOP-8CQAO1L:~/Skill$ gcc skill3.c -o skill3
sadwika@DESKTOP-8CQAO1L:~/Skill$ ./skill3
=== Command History Program ===
Enter commands (type 'exit' to stop)

Command: ls
Command stored successfully.
Command: pwd
Command stored successfully.
Command: history
Command stored successfully.
Command: exit

=== Command History ===
1: ls
2: pwd
3: history

Escape Sequence Demonstration:
New Line -> Hello
World
Tab -> Hello    World
Backslash -> C:\Program Files
sadwika@DESKTOP-8CQAO1L:~/Skill$
```

2.

```
sadwika@DESKTOP-8CQAO1L:~/Skill$ nano skill3dynamic.c
sadwika@DESKTOP-8CQAO1L:~/Skill$ gcc skill3dynamic.c -o skill3dynamic
sadwika@DESKTOP-8CQAO1L:~/Skill$ ./skill3dynamic
=== Dynamic Buffer and Linked List ===
Enter command (type 'exit' to stop): ls
Command stored successfully.
Enter command (type 'exit' to stop): pwd
Command stored successfully.
Enter command (type 'exit' to stop): gcc test.c
Command stored successfully.
Enter command (type 'exit' to stop): exit

=== Command History ===
1: ls
2: pwd
3: gcc test.c

=== Linked List ===
ls
pwd
gcc test.c

Memory released successfully.
sadwika@DESKTOP-8CQAO1L:~/Skill$ nano skill3dynamic.c
```

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GNU nano 8.7.1 skill3dynamic.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Node {
    char *data;
    struct Node *next;
};

int main() {
    int capacity = 2;
    int count = 0;

    char **commands = malloc(capacity * sizeof(char *));

    if (commands == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }

    printf("=== Dynamic Buffer and Linked List ===\n");

    while (1) {
        char buffer[100];

        printf("Enter command (type 'exit' to stop): ");
        if (fgets(buffer, sizeof(buffer), stdin) == NULL) {
            break;
        }

        buffer[strcspn(buffer, "\n")] = '\0';

        if (strcmp(buffer, "exit") == 0)
            break;

        if (count == capacity) {
            capacity *= 2;
```

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GNU nano 8.7.1 skill3dynamic.c

```
            char **temp = realloc(commands,
                                  capacity * sizeof(char *));

            if (temp == NULL) {
                printf("Memory reallocation failed!\n");

                for (int i = 0; i < count; i++) {
                    free(commands[i]);
                }

                free(commands);
                return 1;
            }

            commands = temp;
        }

        commands[count] = malloc(strlen(buffer) + 1);

        if (commands[count] == NULL) {
            printf("Memory allocation failed!\n");

            for (int i = 0; i < count; i++) {
                free(commands[i]);
            }

            free(commands);
            return 1;
        }

        strcpy(commands[count], buffer);
        count++;

        printf("Command stored successfully.\n");
    }

    printf("\n=== Command History ===\n");
```

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GNU nano 8.7.1 skill3dynamic.c

```
    }

    struct Node *head = NULL;
    struct Node *tail = NULL;

    for (int i = 0; i < count; i++) {
        struct Node *newNode = malloc(sizeof(struct Node));

        if (newNode == NULL) {
            printf("Node memory allocation failed!\n");
            break;
        }

        newNode->data = malloc(strlen(commands[i]) + 1);

        if (newNode->data == NULL) {
            free(newNode);
            printf("Node data allocation failed!\n");
            break;
        }

        strcpy(newNode->data, commands[i]);
        newNode->next = NULL;

        if (head == NULL) {
            head = newNode;
            tail = newNode;
        } else {
            tail->next = newNode;
            tail = newNode;
        }
    }

    printf("\n=== Linked List ===\n");

    struct Node *current = head;
```

```
    while (current != NULL) {
        printf("%s\n", current->data);
        current = current->next;
    }

    current = head;

    while (current != NULL) {
        struct Node *nextNode = current->next;
        free(current->data);
        free(current);
        current = nextNode;
    }

    for (int i = 0; i < count; i++) {
        free(commands[i]);
    }

    free(commands);

    printf("\nMemory released successfully.\n");

    return 0;
}
```

OUTPUT:

```
sadwika@DESKTOP-8CQAO1L:~/Skill$ nano skill3dynamic.c
sadwika@DESKTOP-8CQAO1L:~/Skill$ gcc skill3dynamic.c -o skill3dynamic
sadwika@DESKTOP-8CQAO1L:~/Skill$ ./skill3dynamic
=== Dynamic Buffer and Linked List ===
Enter command (type 'exit' to stop): ls
Command stored successfully.
Enter command (type 'exit' to stop): pwd
Command stored successfully.
Enter command (type 'exit' to stop): gcc test.c
Command stored successfully.
Enter command (type 'exit' to stop): exit

=== Command History ===
1: ls
2: pwd
3: gcc test.c

=== Linked List ===
ls
pwd
gcc test.c

Memory released successfully.
sadwika@DESKTOP-8CQAO1L:~/Skill$ nano skill3dynamic.c
```